Harness



File: harness.kicad_sch

Daughterboard



File: daughter.kicad_sch

Wideband, MAP, USB, AUX



File: io.kicad_sch

Miata ECU motherboard 1989–1995 (NA6 / NA8). Carries the year-specific harness header and the year-independent I/O (wideband, MAP, IAT, USB-C). All engine control is on the daughterboard.

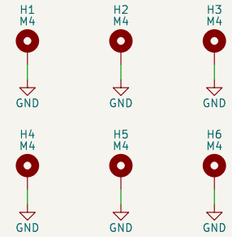
Sheet: /
File: miata_mb_8995.kicad_sch

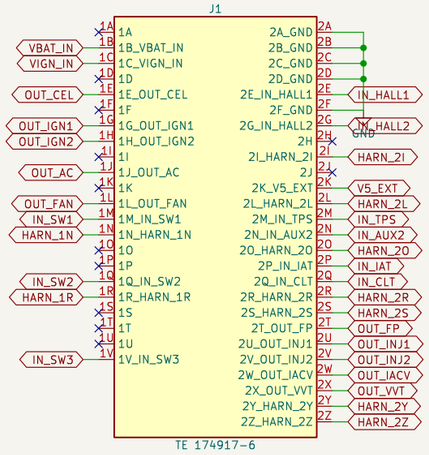
**Title: Miata ECU motherboard 1989–1995 (NA6 / NA8)**

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-30 | **Rev: 0.13** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/4 |

Stock harness header (Miata ECU motherboard 1989–1995): every pin labelled with the daughterboard signal it drives

Case mounting bosses (M4), tied to ground




Generated from scripts/mb_na6.py; netlist identical to the routed board

Sheet: /Harness/
File: harness.kicad_sch

**Title: Miata ECU motherboard 1989–1995 — Harness**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-30 | **Rev: 0.13** |
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## Daughterboard sockets: pin-for-pin with the daughterboard J1 (top) and J2 (bottom)

## Daughterboard standoffs (M3)

HD1 M3 standoff
HD2 M3 standoff
HD3 M3 standoff
HD4 M3 standoff







Generated from scripts/mb_na6.py; netlist identical to the routed board

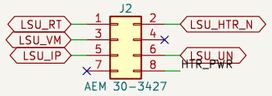
Sheet: /Daughterboard/
File: daughter.kicad_sch

**Title: Miata ECU motherboard 1989–1995 — Daughterboard**

| Size: A3 | Date: 2026-10-30 | **Rev: 0.13** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/4 |

## Wideband sensor harness AEM 30-3427 (controller on the daughterboard)

AEM pin: 1 trim (RT)  2 H−  3 common (VM)  5 pump (IP)  6 sense (UN)  8 H+.  JST PUD S08B-PUDSS-1 (plug PUDP-08V-S)



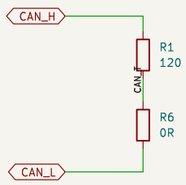




Heater fuse: 2 A hold, 30 V PTC

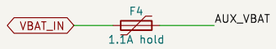
IP/VM/RT/UN and H− run to daughterboard J2 33–38; the heater current returns on J2 39/40

## CAN bus termination at the ECU end

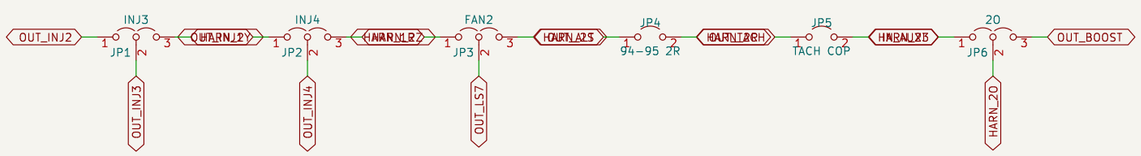


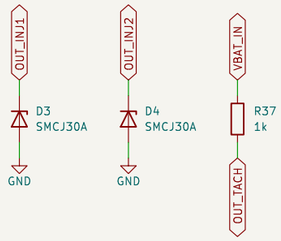

120R; remove R6 if the bus is terminated elsewhere

## Fused +12 V feeds (harness actuators, AUX)

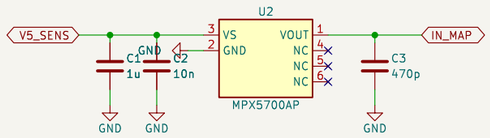



## 1989–95 year / option jumpers, injector clamps (1989–93 paired wires), tachometer pull-up (JP5)

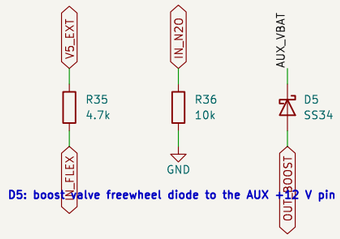



## MAP sensor

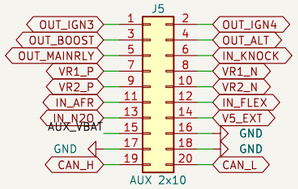



## Flex-fuel sensor pull-up (5 V) and nitrous arming pull-down; both inputs are on the AUX header

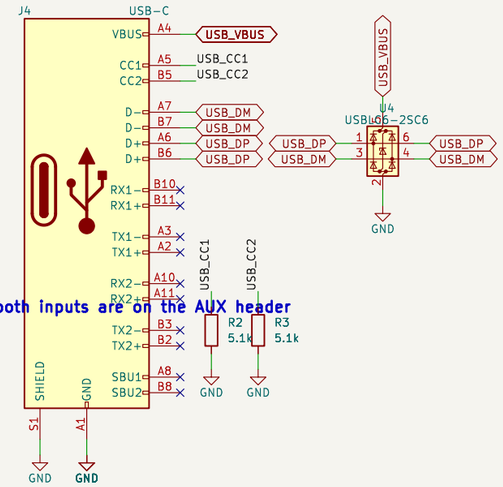



D5: boost-valve freewheel diode to the AUX +12 V pin

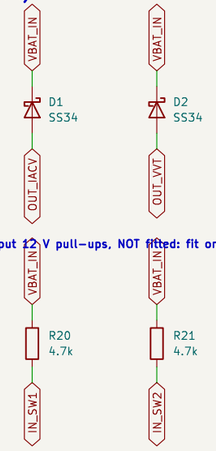
## AUX: add-on I/O the stock harness has no pins for (+12 V through F4)



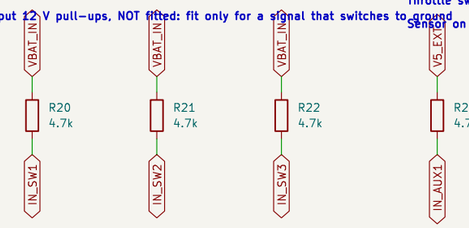

## USB-C (device): tuning, logs, DFU firmware update



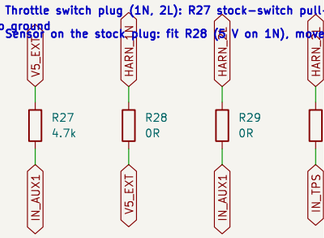

## Valve freewheel diodes (to each valve's +12 V feed) and fit options




Switch-input 12 V pull-ups, NOT fitted: fit only for a signal that switches to ground




Throttle switch plug (1N, 2L): R27 stock-switch pull-up (NOT fitted)
Sensor on the stock plug: fit R28 (5 V on 1N), move R29 to R30




Generated from scripts/mb_na6.py; netlist identical to the routed board

Sheet: /Wideband, MAP, USB, AUX/
File: io.kicad_sch

**Title: Miata ECU motherboard 1989–1995 – Wideband, MAP, USB, AUX**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-30 | **Rev: 0.13** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/4 |